

Homework (Habanero)

- Q1.** Solve for x : $2x + 5 = 11$
- (A) $x = 2$
(B) $x = 3$
(C) $x = 4$
(D) $x = 5$
(E) $x = 6$
- Q2.** A patient is taking a medication that increases in dosage by 3 milligrams each day. If the initial dosage is 10 milligrams, what is the dosage after 4 days?
- (A) $10 + 3 \cdot 4 = 22$ milligrams
(B) $10 + 2 \cdot 4 = 18$ milligrams
(C) $10 + 4 \cdot 3 = 22$ milligrams
(D) $10 - 3 \cdot 4 = -2$ milligrams
(E) $10 - 2 \cdot 4 = 2$ milligrams
- Q3.** Solve for y : $y - 2 = 7$
- (A) $y = 5$
(B) $y = 6$
(C) $y = 9$
(D) $y = 10$
(E) $y = 11$
- Q4.** The number of bacteria in a petri dish doubles every 2 hours. If there are initially 100 bacteria, how many will there be after 6 hours?
- (A) $100 \cdot 2^3 = 800$ bacteria
(B) $100 \cdot 2^2 = 400$ bacteria
(C) $100 \cdot 2^1 = 200$ bacteria
(D) $100 \cdot 2^4 = 1600$ bacteria
(E) $100 \cdot 2^5 = 3200$ bacteria
- Q5.** Solve for z : $z/4 = 9$
- (A) $z = 36$
(B) $z = 40$
(C) $z = 45$
(D) $z = 32$
(E) $z = 28$
- Q6.** A medical researcher is studying the effect of a new drug on blood pressure. The researcher finds that for every 2 milligrams of the drug, blood pressure decreases by 1 point. If a patient's initial blood pressure is 140 points, what dosage will lower their blood pressure to 120 points?
- (A) $2 \cdot (140 - 120) = 40$ milligrams
(B) $2 \cdot (140 - 120) = 20$ milligrams
(C) $1 \cdot (140 - 120) = 20$ milligrams
(D) $2 \cdot (120 - 140) = -40$ milligrams
(E) $1 \cdot (140 - 120) = 20$ milligrams
- Q7.** Solve for x : $x + 2 = 9$
- (A) $x = 5$
(B) $x = 7$
(C) $x = 11$
(D) $x = 13$
(E) $x = 15$
- Q8.** A pharmaceutical company is producing a new medication. The cost of producing x units of the medication is given by $C(x) = 2x + 1000$. What is the cost of producing 250 units?
- (A) $C(250) = 2 \cdot 250 + 1000 = 1500$
(B) $C(250) = 2 \cdot 250 + 1000 = 2000$
(C) $C(250) = 2 \cdot 250 + 1000 = 2500$
(D) $C(250) = 2 \cdot 250 + 1000 = 3000$
(E) $C(250) = 2 \cdot 250 + 1000 = 15000$

Open Ended Questions (FRQ)

- Q9.** Solve for x : $3x - 2 = 2x + 5$. Show your work and check your solution.
- Q10.** The number of cells in a culture grows according to the equation $N(t) = 200 \cdot 2^t$, where t is time in hours. If there are currently 800 cells, how many hours will it take for the culture to reach 1600 cells? Show your work and check your solution.

Chef's Tips

- Read each question carefully
- Check your units and calculations
- Show all work for free response questions